Current DS interview structure

→ Apply → DS/SD = Skills → pandas
→ numpy
→ sk-learn
→ pytorch
→ tensorflow
→ forecasting
→ Analytic
→ sql
→ ....

OA → Link ..... online Q → idea
→ Questions
↳ Human ask?

Assignment → Picky ✗
↳ online Q → Data
↳ code
↳ submit before deadline.

Tech → Python / problem solving.
$\boxed{\text{linus}}$ $\boxed{\text{reddit}}$

ML → depth
↳ breadth / case study.

Breadth → Exp. → Minimum
↳ casestudy. → ML algo

Depth → 1 to 2 algo
(DT)

Temp C → Temp F
↳ LR $F = \frac{9}{5} C + 32$

intercept?

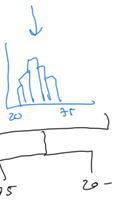
$\hat{y} = w_1 x + \boxed{w_0}$ → intercep

residuals
$= y - \hat{y}$

$\hat{y} > y \Rightarrow < 0$

$\hat{y} < y \Rightarrow$ under predicting residuals

Physics/100	Maths/100	Mechatronics/100
65	68	74
72	67	70
35	40	73



$\Rightarrow$ DT $\Rightarrow$ regression

MAPE → $\boxed{1\%}$

$\frac{1}{n} \sum \frac{(y-\hat{y})}{y}$

DT → output $\Rightarrow$ leaf node average

$\boxed{20-75} \Rightarrow \boxed{20-75}$

% $\Rightarrow$ never ever predict beyond 75%

✓ linear regression

Classification system → $k = \sqrt{\frac{4}{5}}$ → P(+) → dist

x → KNN → $\hat{y}_{knn}$
→ DT → $\hat{y}_{DT}$

$\hat{y}$

(log) $P(+) = \frac{\#+}{\text{Total}\#}$

① Voting:

P1	P2	
1	1	⇒ 1
0	0	⇒ 0
1	0	⇒ ?
0	1	

Prob = (average)

knn → $P - \delta_1$
DT → $P - \delta_2$

weighted average = $w_1\delta_1 + w_2\delta_2$ two
↳ meta model = $\hat{y}$

How can we use SVM for multi-class classification?

One vs rest approach